IP-SceneNet: A Multimodal Network for Scene Classification With Remote Sensing Images and LiDAR Point Clouds

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***Abstract*—The remote sensing scene classification is critical in a wide range of applications. A significant number of studies have adopted deep learning methods for scene classification from remote sensing image datasets. While 2-D information has been fully exploited, 3-D structures of different remote sensing scenes are often overlooked by existing studies, which greatly limits the classification accuracies of scenes, especially in complex urban areas. To tackle this issue, in this article, we present a novel image point scene network (IP-SceneNet) to extract features from the multimodal heterogeneous data of high-spatial resolution (HSR) images and aerial light detection and ranging (LiDAR) point clouds for scene classification. The IP-SceneNet solves three key challenges that occur in multimodal scene classification. It first employs a voxel preprocessing scheme to compress the point clouds to address the information scale imbalance between the two modalities. Second, a dense pooling module is introduced to convert sparse point cloud voxel features into dense representations, mitigating the feature density imbalance. Third, a cross-attention mechanism, termed image-voxel attention (IVA) module, is implemented for the fusion of point cloud and image features, which addresses the problem of intermodality correlations and redundancies. While we were developing IP-SceneNet, two datasets, named 2D3DScene-Netherlands and 2D3DScene-France, were established for tasks of multimodal scene classification. Experiments show that our proposed method achieved overall accuracies of 86.11% and 87.33% on the 2D3DScene-Netherlands and the 2D3DScene-France datasets, significantly outperforming both unimodal and other existing multimodal methods. Further ablation experiments confirmed the significant contributions of HSR images and LiDAR point clouds to scene classifications. In addition, the effectiveness of the dense pooling module and IVA module in IP-SceneNet has also been validated. Our code and the above two datasets are available at: https://github.com/treemanzzz/IP-SceneNet**

***Index Terms*—Data fusion, feature imbalance, multimodal attention, scene classification.**

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I. Introduction

REMOTE sensing scenes pertain to specific areas of the Earth's surface observed and recorded through the remote sensing technology, and these areas typically encompass various geographic entities, such as buildings, trees, and lakes [1]. The classification of remote sensing scenes involves assigning semantic labels to the observed content, enabling a systematic categorization based on varying characteristics. This process finds extensive utility in diverse applications, including environmental monitoring [2], [3], [4] and urban planning [5], [6].

Numerous studies have employed optical remote sensing images for the classification of remote sensing scenes, with methodologies evolved significantly, transitioning from low-level visual features, such as color, texture, and structure [7], [8], [9], [10], and intermediate visual features such as the bag-of-visual words (BOVW) method [11], [12], [13], to advanced deep learning techniques, which have substantially enhanced the accuracy of remote sensing scene classification. However, with the continuous advancements in the sensor technology and the expansion of geographic big data, the volume, quality, and accessibility of multimodal data have significantly improved, facilitating the acquisition of diverse geographic information. Concurrently, the intensification of human activities has rendered remote sensing scenes, particularly urban environments, increasingly complex. Consequently, relying solely on single-modal optical images has become inadequate for accurate classification. In response to this challenge, many studies have incorporated data from other modalities, such as high-spatial resolution (HSR) images [14], [15], [16], [17], [18], [19], synthetic aperture radar (SAR) images [20], [21], [22], [23], multispectral remote sensing images [24], and hyperspectral remote sensing images [25]. Multimodal approaches play a critical role in remote sensing scene classification by leveraging the complementary strengths of different data sources. Nevertheless, the classification methods associated with these multimodal data primarily focus on 2-D images, often neglecting the 3-D structural differences between various remote sensing scenes, which can serve as critical classification criteria.

The incorporation of 3-D information into scene recognition is achieved through the fusion of 2-D and 3-D data. The data fusion can occur either at the data or feature level, or at the decision level [26]. Airborne light detection and ranging (LiDAR) point clouds offer detailed descriptions of

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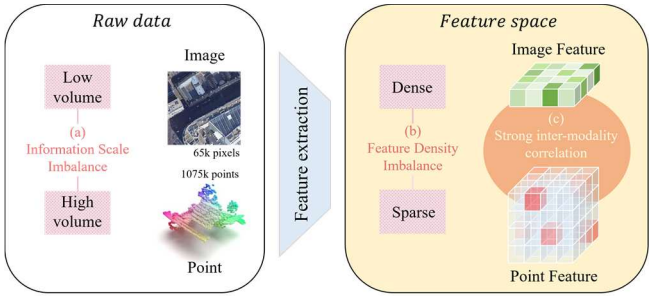


Fig. 1. Illustration of three problems in the process of point cloud and image fusion. (a) Information scale imbalance. (b) Feature density imbalance. (c) Strong intermodality correlation and feature redundancies.

the Earth's 3-D terrain contours, which can be fused with the 2-D data. The previous research has demonstrated the potential of integrating LiDAR point clouds with remote sensing images for tasks, such as ground object detection [27], [28] and semantic segmentation [29], [30]. In the context of remote sensing scene classification, Sanlang et al. [31] extracted spectral and texture features from HSR images, along with digital surface models (DSMs), sky view factors, and flatness from LiDAR point clouds. These 2-D and 3-D features were then separately input into classification models. However, this method's reliance on manually selected low-level visual features limits the provision of high-dimensional information. Al-Najjar et al. [32] converted the LiDAR point cloud data into DSM images and concatenated these with remote sensing images by channels. The combined data were then input into a convolutional neural network (CNN) for feature extraction and classification. However, this approach achieves the fusion of LiDAR point clouds by mapping them onto remote sensing images, leading to information loss from the LiDAR data. The existing research on integrating remote sensing images with LiDAR point cloud classification for remote sensing scenes has not yet achieved end-to-end classification by inputting the original point cloud into deep learning models.

On the other hand, the fusion of LiDAR point clouds with HSR images presents three challenges. First, a notable issue arises from the substantial disparity in data scale between point cloud and image data, particularly in remote sensing scenes with large areas. For instance, considering the point cloud data obtained from the Actueel Hoogtebestand Nederland (AHN) Project of Dutch Water Commission, each data tile covering an area of 31.25 km^2 amounts to approximately 8 GB. In contrast, the data volume of the RGB three-channel HSR image (with a resolution of 0.25 m) within the same coverage area is merely about 80 MB, leading to significant information scale imbalances [33]. This imbalance in scale will lead to a large proportion of the point cloud in the feature space after feature extraction so that the image features cannot be paid attention to. Second, the inconsistent data density of point cloud data, compared with the uniform structure of image data, poses a significant challenge. LiDAR point clouds represent discrete points in a 3-D space, inherently sparse, thus resulting in a sparse distribution of extracted features within the feature space [34]. Conversely, HSR images exhibit dense and continuous characteristics in a 2-D plane, resulting in correspondingly densely distributed extracted features [35]. The fusion of these sparse and dense features introduces challenges related to feature density imbalances [36]. There are a large number of null values in the sparse features of the point cloud, which will make the model unable to pay attention to the valid information of nonnull values when fused with the dense features of the image. Furthermore, there are strong intermodality correlations, which can lead to severe feature redundancy. While the 3-D structural data from point clouds complements the 2-D texture information from images, the richness of these modalities in the feature space may hinder the model's ability to prioritize essential regions, thus reducing the feature recognition efficiency [37]. Consequently, a critical challenge in multimodal fusion tasks is determining how to effectively leverage information from different modalities while minimizing redundancy. The three existing problems mentioned above are illustrated in Fig. 1.

To address the challenges, this study introduces image point scene networks (IP-SceneNets) for the integration of point clouds and images to classify remote sensing scenes. On the one hand, this approach aims to overcome the issue of imbalanced multimodal information and feature fusion when combining point clouds and images. On the other hand, by leveraging the inherent correlations between different modalities, IP-SceneNet can focus on the most important regions of different modalities and utilize them to alleviate the problem of feature redundancy. Consequently, this method facilitates the precise classification of remote sensing scenes by enhancing feature expression through the synergistic fusion of point clouds and images. To further validate the effectiveness of the proposed IP-SceneNet and address the gap in current remote sensing scene classification datasets, which lack 3-D point cloud information, this study has developed two new datasets: the 2-D–3-D Scene Dataset in the Netherlands (2D3DScene-Netherlands) and the 2-D–3-D Scene Dataset in France (2D3DScene-France). Both datasets include point clouds and images for each remote sensing scene category, providing a comprehensive foundation for more accurate and robust scene classification.

The main contributions of this article can be summarized in the following four points.

1) The study introduced two novel sets of remote sensing scene datasets, 2D3DScene-Netherlands and 2D3DScene-France. These datasets are unique as they combine HSR images with LiDAR point clouds, representing the first datasets to incorporate point cloud data for remote sensing scene classification.
2) The development of an end-to-end deep learning model, IP-SceneNet. This model employs a dual-branch architecture to extract features from distinct modalities and enables efficient cross-modal interaction and fusion. Through detailed comparative and ablation experiments on 2D3DScene-Netherlands and 2D3DScene-France datasets, the effectiveness and logical flow of IP-SceneNet are demonstrated.
3) The design of dense pooling, a simple yet effective pooling structure. This structure facilitates the conversion

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of sparse features to dense features, thereby reconciling the disparity in feature density between point cloud and image data.

4) The design of a multimodal attention module, i.e., image-voxel attention (IVA) module, for the fusion and interaction of point cloud and image. This model employs a multimodal attention mechanism to compute the correlation between different modalities, concentrating on features from highly correlated regions while automatically filtering out features from low-correlation regions. This approach achieves feature screening and enhancement, effectively reducing the feature redundancy.

While the multimodal fusion of optical imagery and LiDAR point clouds can substantially enhance the scene classification, it is important to note that not all application scenarios require fused classification. Our study positions multimodal fusion as a complementary strategy that becomes particularly valuable in complex and heterogeneous environments, where single modalities are insufficient.

II. Related Works

A. Deep-Learning-Based Methods for Scene Classification

CNN was the first deep learning approach adopted for remote sensing scene classification, gaining widespread application in 2015 [9]. Compared with traditional advanced methods, CNN offers the advantage of end-to-end feature learning and can automatically extract high-level visual features that are beyond the reach of conventional techniques [38]. Various CNN-based remote sensing scene classification methods have been developed, utilizing different strategies, including using pretrained CNN as feature extractor [39], [40], [41], fine-tuning pretrained CNN on target datasets [42], [43], [44], and training CNN from scratch [32], [45], [46], [47].

While CNNs excel at processing local information, they struggle to fully capture global features. Vision Transformer (ViT) is well-suited for learning long-range contextual relationships. Bazi et al. [48] introduced ViT for remote sensing scene classification and achieved superior results on multiple datasets compared with the most advanced CNN-based methods. Xu et al. [49] proposed ET-GSNet, integrating ResNet-18 as the student model and ViT as the teacher model, utilizing the knowledge distillation to combine the strengths of both models. Guo et al. [50] developed a channel–spatial attention transformer (CSAT), combining the channel–spatial attention (CSA) mechanism with the ViT method, to create an effective HSR image scene classification framework. The CSAT model effectively captures and preserves both local and global knowledge in remote sensing scenes.

In addition, some researchers have turned to adversarial generative networks (GANs) to address the issue of insufficient labeled samples for remote sensing scene classification. Lin et al. [51] introduced the multiple-layer feature-matching GAN (MARTA GAN), which fuses mid-level features with global features to learn better representations through feature descriptors. Ma et al. [52] proposed SiftingGAN, an innovative method that extends the traditional GAN framework to generate a larger, more diverse, and more realistic set of labeled samples for data augmentation so that the generated samples can be applied to other deep learning networks.

B. Scene Identification With Multiple Data Modalities

While the majority of scene classification leverages high-resolution remote sensing imagery, numerous studies have also explored the fusion of multiple data modalities to achieve more precise and comprehensive scene recognition. These classification methodologies can be categorized into two types: homogeneous fusion and heterogeneous fusion.

The homogeneous fusion involves the use of different data modalities in the form of traditional spectral images, such as HSR images, multispectral/hyperspectral remote sensing images, and nighttime light images. For instance, Zhu et al. [24] utilized IKONOS sensors to collect panchromatic and multispectral images over Wuhan, China. They applied a multiobject spatial context relationship model, a topic model, and Google LeNet to this dataset, demonstrating that fusion classification methods outperformed others. Similarly, Huang et al. [53] fused multiview optical images with high-resolution nighttime light images to recognize and classify urban functional areas in Wuhan and Beijing, China.

With advancements in sensor and information technology, data imaging and acquisition methods have diversified, leading to the development of heterogeneous fusion methods in remote sensing. Heterogeneous fusion integrates data from different imaging mechanisms and structures, such as SAR images, LiDAR point clouds, and geospatial big data. Wang et al. [54] proposed an adversarial network, S-CycleGAN, capable of generating optical images from SAR images. This method is particularly useful when optical remote sensing data are unavailable, providing alternative data sources. They released a paired SAR–optical image dataset covering 32 cities in China based on this approach. In addition, Su et al. [55] developed a universal urban scene understanding framework (USUMG) that integrates multiple geographic data sources, including HSR images, OpenStreetMap (OSM) road network data, and points of interest (POIs) data. This framework combines high-level semantic features from high-resolution remote sensing images and socioeconomic features from POI data to categorize urban scenes.

Despite the success in integrating various data modalities to enhance the remote sensing scene analysis and classification accuracy, the current research has not yet delved into the 3-D structure of internal features in remote sensing scenes. There is a notable gap in the development of relevant datasets and foundational models for this purpose.

C. Multimodal Learning With Images and Point Clouds

The complementary information embedded in multimodal data can have a more efficient and robust effect on the task and can obtain more accurate and better results than single-modal data [31], [56], [57]. Therefore, the fusion of image and point cloud for classification and detection is becoming a research hotspot and shows great potential. Existing methods for the fusion of point clouds and images can be categorized

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into three primary approaches: mapping, intermodal guidance, and result aggregation, corresponding to early fusion, middle fusion, and late fusion methods.

Mapping aims to bridge the structural gap between the two modalities by transforming 3-D point cloud data into 2-D image space or projecting images onto LiDAR point clouds. The former case yields products such as DSM, bird's eye view (BEV) or intensity images, which are then used to extract 2-D features for subsequent analysis and classification. Hosseinpour et al. [58] proposed a gated fusion network, CMGFNet, capable of achieving cross-modal and multilevel feature fusion between HSR images and DSMs derived from point clouds. Meyer et al. [59] correlated LiDAR point clouds with camera pixels by projecting the 3-D data onto 2-D images, subsequently feeding the mapping results into a CNN for feature extraction, which enabled the semantic segmentation based on the extracted features. Similarly, Wang et al. [33] developed a dimension sensor structure to align point cloud features and image features at the feature level, resulting in a higher degree of correspondence between the two modalities. While these methods effectively achieve cross-modal learning and demonstrate improved performance over single-modal approaches, the process of mapping 3-D data to 2-D inevitably leads to the loss and distortion of crucial features, particularly geometric structure information. In the latter case, HSR images are projected onto 3-D point clouds to form colored points. In [60], HSR images and point clouds were first co-registered, and spectral features of images were transferred to points. Finally, the multispectral points were input to a multiscale PointNet [61] for semantic segmentation.

The intermodal guidance emphasizes the dominance of one modality, with auxiliary information provided by other modalities. The fusion process involves one modality guiding the others. For instance, Braun et al. [62] focused on 2-D detection by utilizing stixels generated from 3-D LiDAR data to guide object detection in 2-D images, leading to further refinement. Similarly, Ku et al. [63] employed LiDAR point clouds to predict 3-D candidate regions, which were then combined with RGB images to generate multiview images of the target. Features of these multiview images were subsequently used to correct prior detection results. Wang et al. [33] introduced the concepts of "weak modality" and "strong modality," arguing that in modal fusion and LiDAR point clouds, which contain richer information, should be considered the strong modality. By doing so, the image information was supplemented, thus enhancing the semantic segmentation. While the intermodal guidance offers flexibility, diversity, and adaptability, practical applications often lead to the dominance of a single modality, which may not be suitable for all categories. Some categories may depend more on information from another modality, resulting in suboptimal classification performance.

Result aggregation involves separate processing of LiDAR data and image data to obtain distinct results in target recognition, object detection, or semantic segmentation, which are further aligned. This alignment involves matching the 2-D and 3-D results for the same target, determining their spatial positions and features, and subsequently fusing these results using various methods. For instance, Pang et al. [64] utilized a 2-D detection model for RGB images and a 3-D detection model for LiDAR point clouds. They integrated multiple statistical features, such as confidence, distance, and intersection over union (IoU), to project the 3-D bounding box onto the 2-D image accurately. Similarly, Hang et al. [65] employed two coupled CNNs: one to learn spectral–spatial features from hyperspectral data and the other to capture elevation information from LiDAR data. They applied a weighted sum strategy to fuse the classification results, where the weight is determined by the classification accuracy of each output. The method of result aggregation does not require feature fusion during the feature extraction process, which limits the ability to leverage the interdependence between different modalities. In addition, this approach may require multiple classifiers or detectors, leading to inefficient computational resource usage.

III. Datasets

We have compiled two distinct datasets comprising both HSR images and LiDAR point clouds. One is the 2D3DScene-Netherlands, which was collected in Amsterdam and The Hague, The Netherlands. The other is the 2D3DScene-France, which was collected in Montpellier, Lyon, and Marseille, France.

The HSR images utilized in both datasets were sourced from Google Earth and captured at a resolution of 0.5 m. All the images were cropped and resampled to 256×256 pixels. The LiDAR point cloud data in the 2D3DScene-Netherlands dataset originated from the 4th version of Actueel Hoogtebestand Nederland (AHN-4) Project (https://www.ahn.nl/) conducted by Dutch Water Commission. The AHN-4 project utilized the most advanced airborne LiDAR sensors, such as RIEGL LMS-Q1560, for point cloud data collection in 2020, resulting in an approximate point density ranging between 6 and 10 points per square meter. Meanwhile, the LiDAR point cloud data in the 2D3DScene-France dataset were derived from the LIDAR HD Project (https://geoservices.ign.fr/lidarhd), initiated and led by French National Geographic Forestry Information Institute. The LIDAR HD Project utilized the most advanced sensors, such as RIEGL VQ-1560II and Leica ALS80, for point cloud collection in 2021, resulting in an average point density ranging from 10 to 20 points/m^2. It is important to note that this study exclusively employs the x–y–z coordinates of the point cloud data, while omitting other attributes such as intensity. The 2D3DScene-Netherlands and the 2D3DScene-France datasets were collected over an area of 318.4 and 344.75 km^2, respectively. Moreover, co-registration between LiDAR data and HSR images was conducted by first automatically detecting correspondence points from them, and second calculating 3-D transformation parameters. Based on the computed 3-D transformation parameters, the point clouds were reprojected, and a registration accuracy of less than 1 cm was achieved.

Each of the two datasets encompasses nine distinctive categories that exemplify variances in both 2-D and 3-D representations, including transport (TR), industrial area (IA), office building (OB), sport field (SF), farmland (FA), forest (FO), residential area (RA), church (CH), and meadow (ME),

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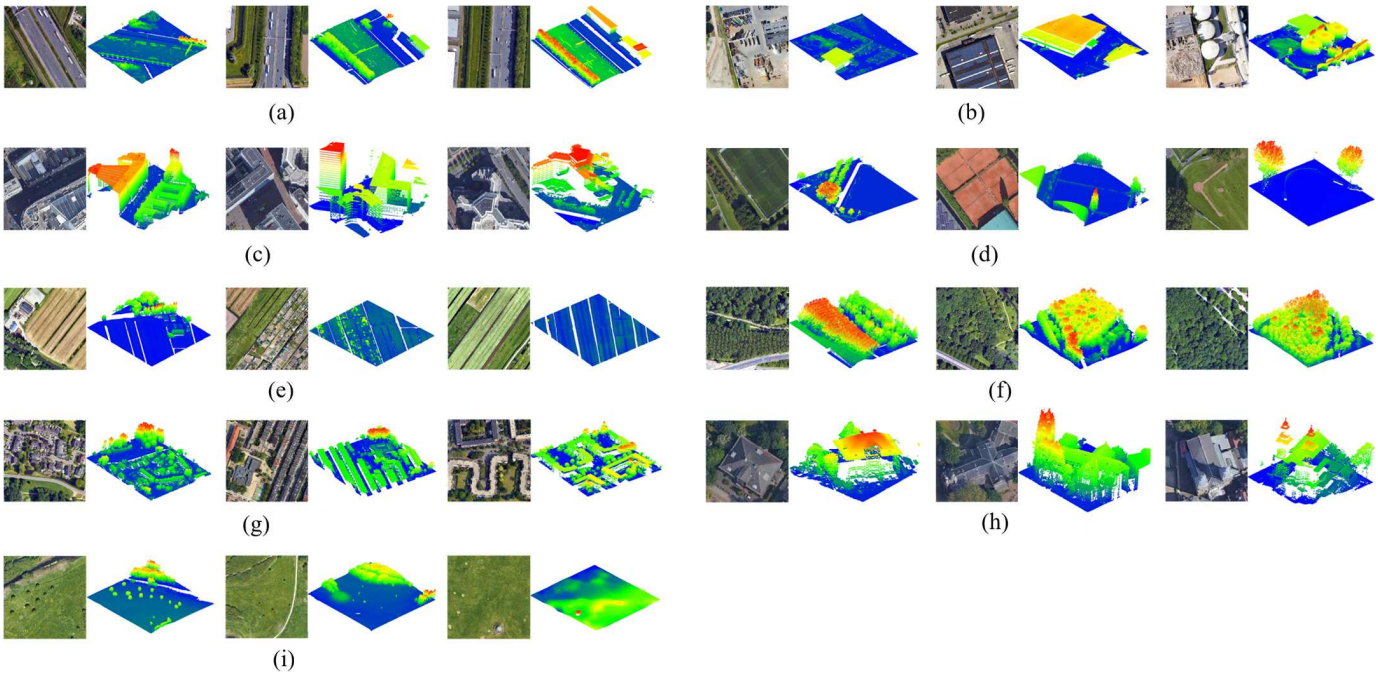


Fig. 2. Sample images and point clouds in the 2D3D-Netherlands dataset. (a) TR. (b) IA. (c) OB. (d) SF. (e) FA. (f) FO. (g) RA. (h) CH. (i) ME.

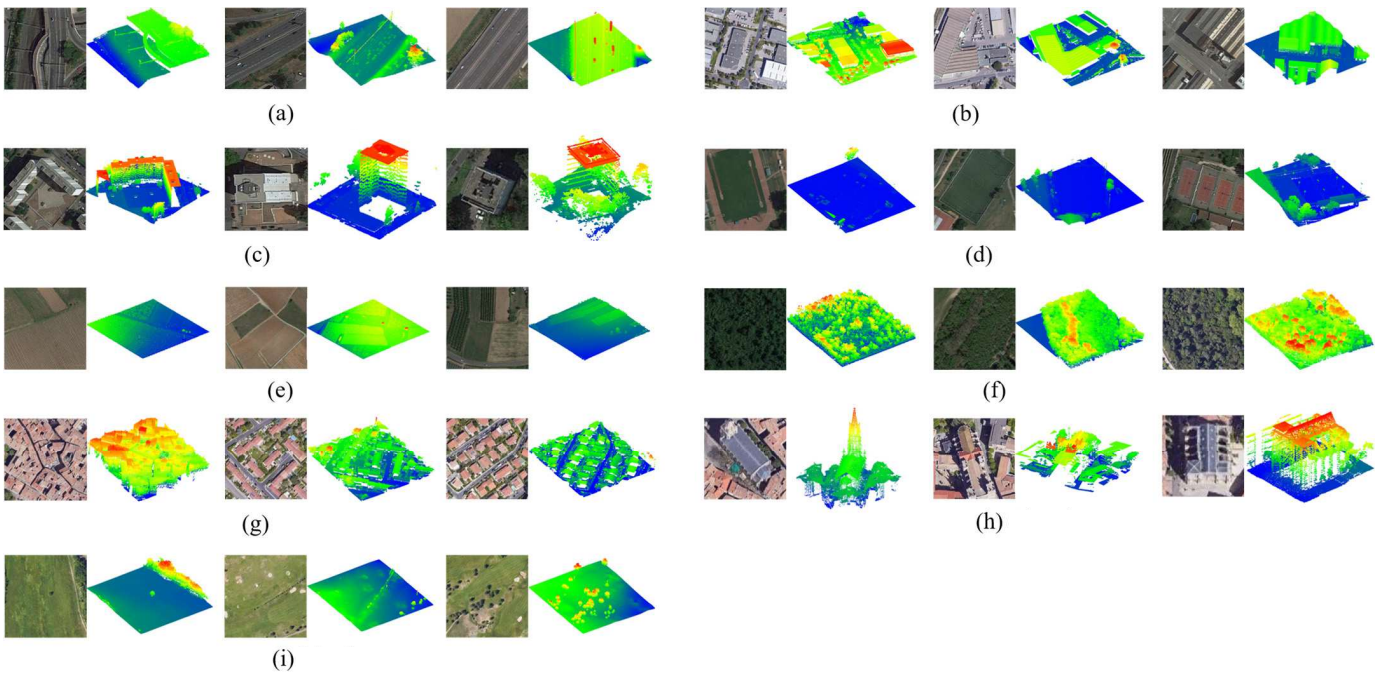


Fig. 3. Sample images and point clouds in the 2D3D-France dataset. (a) TR. (b) IA. (c) OB. (d) SF. (e) FA. (f) FO. (g) RA. (h) CH. (i) ME.

and their information is described in Table I. Categories are manually labeled through expert visual interpretation, utilizing both image and point cloud data. In addition, the POI and area of interest (AOI) information of Google Maps is referenced to verify labeling accuracy. Figs. 2 and 3 present a graphical illustration depicting exemplars belonging to the distinct categories in the two datasets.

In both datasets, we have labeled 250 samples per category, and these samples are distributed over the whole study area. Such a design fosters equilibrium among categories and ensures uniformity in sample sizes for comprehensive analysis and evaluations. All these samples have been uploaded to: https://www.scidb.cn/s/iEzamy

IV. Methodology

A. Overall Architecture of IP-SceneNet

Fig. 4 presents the innovative IP-SceneNet architecture, an end-to-end deep learning network consisting of point cloud and image branches. First, the input point cloud is voxelized

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TABLE I
DEFINITIONS OF THE NINE TYPES OF REMOTE SENSING SCENES IN THIS STUDY

Types	Definitions
Transport (**TR**)	Transportation routes, including roads, railways, bridges, and overpasses.
Industrial Area (**IA**)	Areas for industrial production and storage, such as factory buildings, logistics warehouses, and oil storage tanks.
Office Building (**OB**)	Areas for professional commercial offices, entertainment, and gatherings.
Sport Field (**SF**)	Areas for human sports or athletic competitions.
Farmland (**FA**)	Areas for agricultural production.
Forest (**FO**)	Areas where woody plants are the main biome.
Residential Area (**RA**)	Areas where residential buildings are the main body and equipped with commercial outlets, cultural and educational, entertainment, greening, public and public facilities.
Church (**CH**)	Areas where Christianity celebrates Mass and other religious matters.
Meadow (**ME**)	Areas where herbaceous and shrub plants predominate.

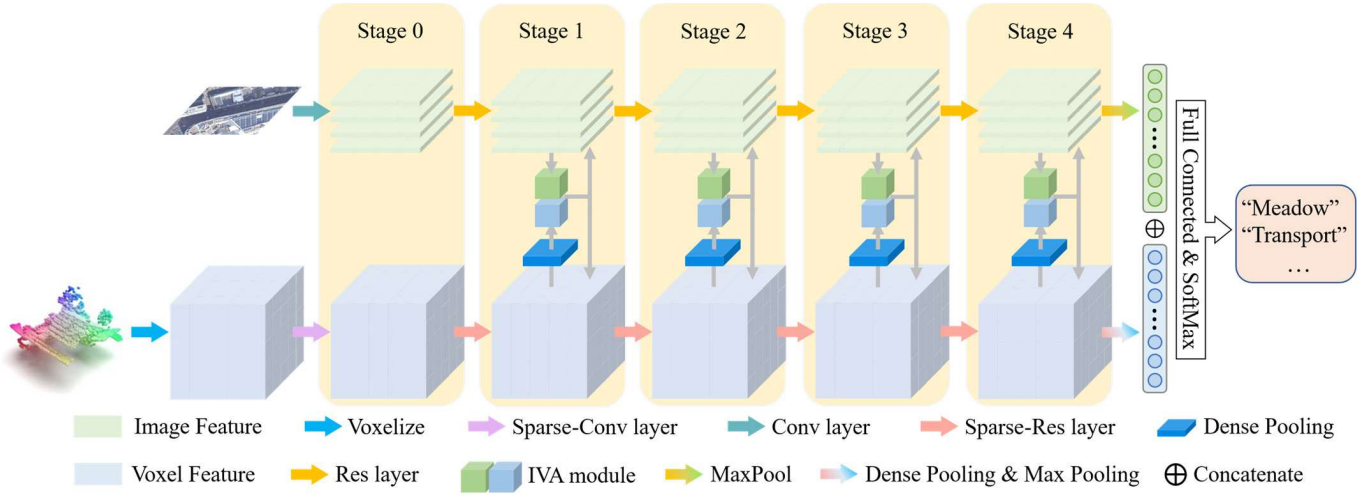


Fig. 4. Overall structure of IP-SceneNet.

before feature extraction. Second, the point cloud and image branches concurrently extract heterogeneous features from their respective modalities. In the image branch, a CNN with a five-stage residual structure is employed to process HSR images and extract pertinent features. Simultaneously, a symmetrical structure of the image branch is adopted in the point cloud branch, where sparse convolutions are utilized to extract point cloud features from the voxels above. Third, feature interactions and fusions of both modalities are performed at various stages in a hierarchical way. A dense pooling module is initially created to adjust the feature density of point clouds to images, thereby mitigating the issue of information density imbalance. Subsequently, a multimodal attention module, named IVA module, considering the correlations between images and point clouds, is proposed to fuse the image and point features effectively. The IVA modules are positioned between the two branches in the last four stages so that the multimodal features can be fused hierarchically. Fourth, multimodal remote sensing scene classification is conducted with the fused features, and classes of remote sensing scenes are output.

B. LiDAR Point Clouds Voxelization

To lower the data volume as well as mitigate the issue of data amount imbalance, voxelization is performed on the original LiDAR point cloud. The voxelization process not only removes the redundant information but also transforms the unstructured point cloud into regular voxels. The whole process can be divided into two steps.

First, the original point clouds in each scene are normalized, and the x, y, and z coordinates of each point are transformed to $[-1, 1]$. This step is crucial as it can prevent issues such as gradient explosion or vanishing gradients in the subsequent neural network training process. Moreover, the normalized data exhibit improved numerical characteristics, which can

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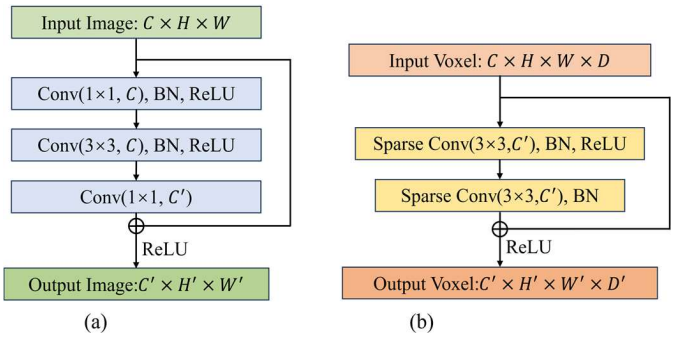


Fig. 5. Structural diagram of the image branch and point cloud voxel branch in a stage. (a) Res layer structure in the image branch. (b) Sparse-Res layer structure in the point cloud voxel branch.

facilitate faster convergence of models toward optimal solutions [61]. The point normalization can be expressed as follows:

$$(\bar{x}_i, \bar{y}_i, \bar{z}_i) = \frac{(x_i, y_i, z_i) - (\bar{x}, \bar{y}, \bar{z})}{r} \tag{1}$$

where (x_i, y_i, z_i) are the coordinates of the ith point, $(\bar{x}, \bar{y}, \bar{z})$ are the coordinate of the centroid point, r is the maximum distance between the centroid point and all points in the scene, and $(\bar{x}_i, \bar{y}_i, \bar{z}_i)$ are the normalized coordinate of the ith point.

Second, after point cloud normalization, all points are bounded by a cube with an edge length of 2 and a centroid point of $(0, 0, 0)$. The cube is equally divided into $512 \times 512 \times 512$ voxels. We chose the size of 512 as it is an integer multiple of the size of the image feature map, which is 256×256. Such a setting is important for feature alignment during the feature fusion process in Section IV-D.

It is important to note that voxelization significantly reduces the information scale of point cloud data. In the case of the 2D3D-Netherlands dataset, the original image data occupied 1.4 GB, while the point cloud data, before voxelization, was substantially larger at 130 GB. However, after voxelization, the size of the point cloud data was reduced to 4.4 GB. This reduction markedly decreases the disparity in information scale between the point cloud and image data.

C. Dual-Branch Networks for Image and Point Hierarchical Feature Extraction of Point Cloud Voxels and Images

A dual-branch network structure is employed to separately extract features from images and point cloud voxels. The dual-branch design allows each modality to be processed in its natural feature space, preserving the modality-specific characteristics before fusion. Compared with early fusion, this structure enables more effective extraction of complementary information from heterogeneous inputs, as shown in prior multimodal literature [33], [59]. The two branches are symmetrical, and each branch utilizes a residual network comprising five stages, as depicted in Fig. 5. The residual structure allows the deep network to have more layers without the issue of gradient explosion and enhances the capacity of feature extraction [66]. Simultaneously, the multistaged network captures features at various levels and resolutions, allowing for multilevel feature extraction and hierarchical feature fusion in the next step [67].

1) Image Feature Extraction: Image patches with a fixed size of $3 \times 256 \times 256$ are input to the image branch, which is based on a residual network and consists of five stages. The first stage, i.e., Stage 0, comprises a 7×7 convolutional layer and a 3×3 max-pooling layer. The subsequent four stages have similar structures, with each stage containing 3, 4, 6, and 3 residual blocks, respectively. The structure of a residual block is shown in Fig. 5(a). Image features are first input to a 1×1 convolutional layer and a 3×3 convolutional layer. We put a batch normalization (BN) layer after each convolutional layer, which can effectively mitigate over-fitting by normalizing features and reducing internal variances between samples. A rectified linear activation function (ReLU) is added to improve the stability of the model. Then, a 1×1 convolutional layer is followed to change the dimensions of the output features. A skip connection and a pixelwise addition are performed on the input and output image features.

Assuming that the output feature map following feature extraction at each stage is represented as $\mathbf{I}_i (i = 0, 1, 2, 3, 4)$. After stage 0, the size of the feature map $\mathbf{I}_0$ is reduced to a quarter of the original dimensions, resulting in a size of $64 \times 64 \times 64$. From stages 1 to 4, the output feature maps generated from feature extraction are subsequently fed into the IVA module. Here, they interact and fuse with the point cloud features, producing new image feature maps, denoted by $\mathbf{RI}_i (i = 1, 2, 3, 4)$. The $\mathbf{RI}_i$ output from the IVA module of each stage serves as the input for the feature extraction process of the next stage. This progress can be expressed as (IFE and IVA represent the image feature extraction operation and IVA module, respectively)

$$\begin{cases} \mathbf{RI}_i = \mathrm{IVA}(\mathbf{I}_i) \\ \mathbf{I}_{i+1} = \mathrm{IFE}(\mathbf{RI}_i). \end{cases} \tag{2}$$

Each stage of feature extraction will change the size of the image feature maps as well as their dimensions, i.e., $256 \times 64 \times 64$ for stage 1, $512 \times 32 \times 32$ for stage 2, $1024 \times 16 \times 16$ for stage 3, and $2048 \times 8 \times 8$ for stage 4. Notably, the IVA module maintains the size of the feature maps throughout these stages.

2) Point Cloud Voxel Feature Extraction (PVFE): After transforming the original point clouds into voxels, point features are extracted from these voxels via the point branch. Symmetrical to the image branch, the point branch also comprises five stages, with each stage corresponding to one stage in the image branch. After point voxelization, the unstructured points become regular voxels, which are more convenient to be processed by 3-D CNNs. However, a large number of voxels are empty, containing zero points. To handle the empty voxels and further decrease the computational amount, sparse convolutional networks are employed in this study. Existing studies have proved that sparse convolutions, which perform convolutional operations exclusively on nonzero or valid data points, can prevent redundant feature extraction and improve network efficiency [68]. This approach is implemented by indexing the locations of nonzero elements and applying convolutions only at these positions and their neighboring regions.

In Stage 0, there are two 2×2 sparse convolutions, which are followed by a 3×3 sparse convolution. A BN layer and

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an ReLU layer are added to a sparse convolutional layer to enhance its stability. In the next four stages, voxel features are first input to a 2×2 sparse convolutional layer followed by a BN layer and an ReLU layer. The output features are then entered into several repeated block units with each one containing sparse convolutions and a residual-like structure [69], [70]. Therefore, we name each block unit a Sparse-Res block. The structure of a Sparse-Res block is shown in Fig. 5(b). Each Sparse-Res block consists of two 2×2 sparse convolutions and BN layers, and an ReLU layer is added to the second sparse convolutional layer. A skip connection and a voxelwise addition are performed on the input and the output of the Sparse-Res block. The last four stages contain two Sparse-Res blocks, individually.

The dimensional size of the original input voxel feature is $\mathbf{V} \in \mathbb{R}^{3\times512\times512\times512}$. After Stage 0, the spatial sizes are reduced to one-quarter of the input, i.e., $128 \times 128 \times 128$. Similar to the image branch, in stages 1–4, the features $\mathbf{V}_i (i = 1, 2, 3, 4)$ obtained from the point cloud branch will be input into the IVA module for interactive fusion with image features. The resulting fused features $\mathbf{RV}_i (i = 1, 2, 3, 4)$ will serve as input for the subsequent stage of feature extraction. This progress can be expressed as (PVFE and IVA represent the point cloud voxel feature extraction operation and IVA module, respectively)

$$\begin{cases} \mathbf{RV}_i = \mathrm{IVA}\left(\mathbf{V}_i\right) \\ \mathbf{V}_{i+1} = \mathrm{PVFE}\left(\mathbf{RV}_i\right). \end{cases} \tag{3}$$

In these stages, the shapes of the point cloud voxel features after extraction are as follows: $32 \times 64 \times 64 \times 64$ for stage 1, $64 \times 32 \times 32 \times 32$ for stage 2, $128 \times 16 \times 16 \times 16$ for stage 3, and $256 \times 8 \times 8 \times 8$ for stage 4. Notably, the IVA module does not alter the dimensions of the point cloud voxel features.

D. Multimodal Feature Interaction and Fusion

Image features capture spectral information and horizontal distributions of ground objects. Conversely, point features capture the height profile and 3-D structural characteristics of the ground terrain. As the two features complement each other greatly, it is important to interact and fuse them for a more comprehensive and better interpretation of ground terrains. In this study, a dense pooling module is adopted to uniformize the density of image and point features, while the IVA module is proposed for the fusion of the multimodal features.

1) Dense Pooling Module: On the one hand, the output voxel features in the point branch are sparse in the 3-D space, with features missing in empty voxels. On the other hand, the output features in the image branch are dense and contiguous in 2-D space. The two features cannot be directly fused as they will inevitably lead to imbalanced information at different locations, adding challenges to deep networks to allocate appropriate attention for each element in space. Therefore, the dense pooling module is proposed and conducted on the voxel features before they are integrated with image features.

The process of dense pooling is illustrated in Fig. 6. A max-pooling operation was conducted along the Z-axis. It divides the sparse features along the Z-axis into a set of columns. For each column, the maximum value is output. The successes of the dense pooling module depend on two factors: the maximum pooling operation and the Z-axis as the pooling direction. First, the maximum pooling operation fully accentuates the salient characteristics of an area [71]. Second, the Z-axis is opted for as it can offer the most comprehensive information in all directions, while other viewpoints may encounter serious obstruction issues [72]. Discussions of pooling directions can further refer to Section VI-B1.

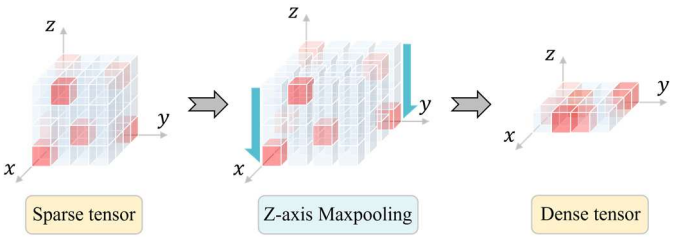


Fig. 6. Specific steps for dense pooling operations.

The outputs of the dense pooling module are dense tensors in 2-D form. In a word, the dense pooling module condenses the voxel features and discards a great deal of redundant features, including those of the null voxels, rendering the point cloud features denser and more contiguous. The dense pooling module will efficiently mitigate the issue of information density imbalance between images and point cloud voxels.

2) IVA Module: The IVA module is designed to leverage a multimodal attention mechanism that enables the joint processing of information from multiple modalities. This mechanism allows the model to receive and integrate inputs from diverse sources simultaneously. Unlike simple concatenation or additive fusion, the attention-based strategy allows each modality to dynamically attend to relevant features from others. As a result, the fused features are enriched not only with modality-specific information but also with cross-modal contextual cues, thereby enhancing the model's ability to capture complex intermodal relationships and semantic alignments.

After the dense pooling operation, the IVA module is introduced and placed between each stage in the image branch and the dense pooling module for the multimodal interaction and fusion of image and point cloud features. The structure of the IVA module is illustrated in Fig. 7. Assume that in the ith stage after feature extraction in Section IV-C, the sparse voxel feature before dense pooling operation can be represented by $\mathbf{V_i} \in \mathbb{R}^{C\times H\times W\times D}$, the dense point cloud features after the dense pooling module are represented by $\mathbf{DV_i} \in \mathbb{R}^{C\times H\times W}$, the image features are represented by $\mathbf{I_i} \in \mathbb{R}^{C\times H\times W}$. A 1×1 convolution is performed on image features $\mathbf{I_i}$, and a reshape and transpose operation is further conducted, which outputs $\mathbf{W_Q}$

$$\mathbf{W_Q} = \left(\mathrm{reshape}\left(\mathrm{conv}_1^{1\times1}\left(\mathbf{I_i}\right)\right)\right)^{\mathrm{T}}. \tag{4}$$

Simultaneously, another 1×1 convolution is performed on the point cloud feature $\mathbf{DV_i}$, which is followed by a reshape operation. The output point cloud feature map is $\mathbf{W_K}$

$$\mathbf{W_K} = \mathrm{reshape}\left(\mathrm{conv}_2^{1\times1}\left(\mathbf{DV_i}\right)\right). \tag{5}$$

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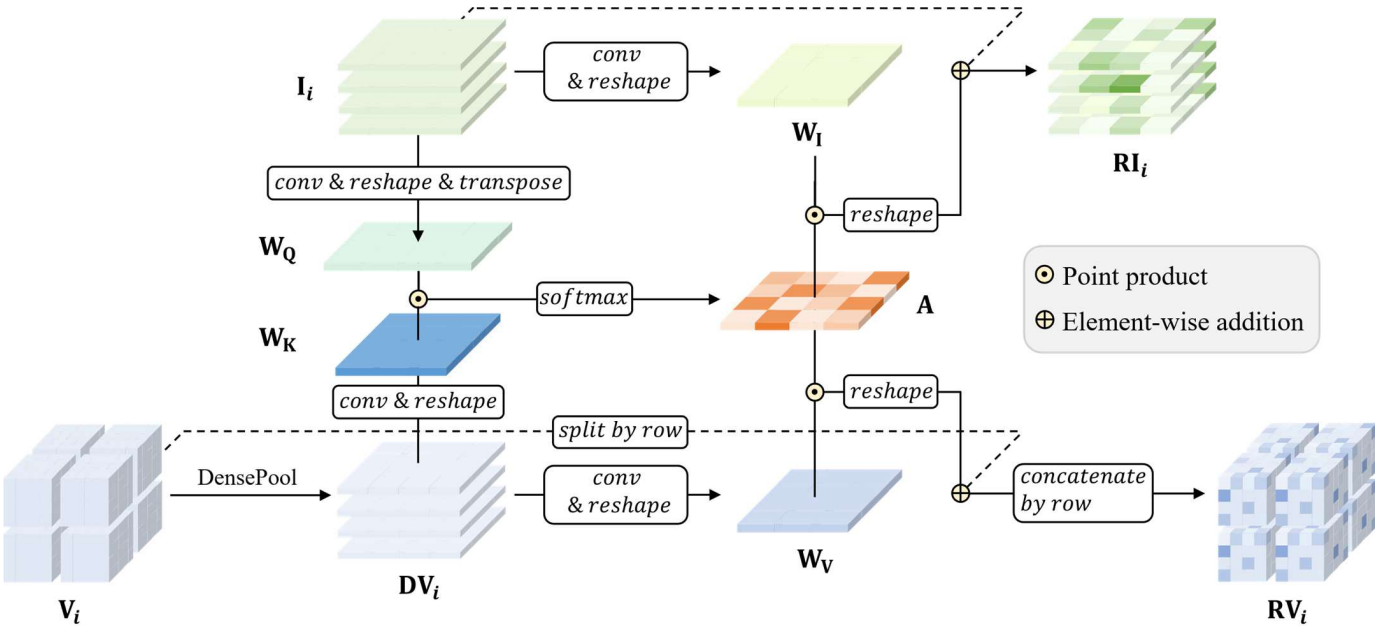


Fig. 7. Working principle of the IVA module.

The matrices $\mathbf{W_Q}$ and $\mathbf{W_K}$ serve as intermediary structures for assessing the similarity between image features and point cloud voxel features. Specifically, they facilitate the mapping of image features and point cloud voxel features into 2-D matrices, respectively.

Then, the dot product of $\mathbf{W_Q}$ and $\mathbf{W_K}$ is computed to produce an attention score matrix, with each element indicating the level of correlation between point clouds and the remote sensing images. A high score in the attention score matrix corresponds to a large correlation between the multimodal features, meaning that greater attention should be paid to this area by our proposed model. Following this, a SoftMax function is employed to normalize the score matrix, generating an attention weight map, denoted by $\mathbf{A} \in \mathbb{R}^{HW \times HW}$

$$\mathbf{A} = \text{softmax}\left(\mathbf{W_Q}\mathbf{W_K}\right). \tag{6}$$

Subsequently, the image feature $\mathbf{I}$ and point cloud feature $\mathbf{DV}_i$ are fed into another two 1-D convolutional layers, followed by reshape operations. Two feature maps $\mathbf{W_I}$ and $\mathbf{W_V}$ are thus generated

$$\mathbf{W_I} = \text{reshape}\left(\text{conv}_3^{1\times1}\left(\mathbf{I}_i\right)\right), \quad \mathbf{W_V} = \text{reshape}\left(\text{conv}_4^{1\times1}\left(\mathbf{DV}_i\right)\right) \tag{7}$$

where $\mathbf{W_I}$ and $\mathbf{W_V}$ represent image features and point cloud voxel features, respectively. Later, the attention weight map $\mathbf{A}$ is elementwise multiplied with $\mathbf{W_I}$ and $\mathbf{W_V}$, separately. After elementwise multiplication, features with high attention weights in both the image and point cloud are emphasized, while irrelevant information is suppressed. This enables the model to automatically disregard redundant features, focusing on more relevant features. This process not only improves feature interactivity but also strengthens the model's ability to fuse multimodal features. Finally, the resultant feature maps are elementwise added to the input features via a residual connection. The output features are represented as $\mathbf{RI}_i$ and $\mathbf{RV}_i$

$$\mathbf{RI}_i = \text{reshape}\left(\mathbf{A}\cdot\mathbf{W_I}\right) + \mathbf{I}_i, \quad \mathbf{RV}_i = \text{reshape}\left(\mathbf{A}\cdot\mathbf{W_V}\right) + \mathbf{V}_i. \tag{8}$$

Consequently, the IVA module leverages the correlations between multimodal features for feature interaction, offering mutual guidance and facilitating effective feature fusion.

E. Multimodal Remote Sensing Scene Classification

The output image features ($\mathbf{RI_4}$) from Stage 4 in the image branch have a dimensional size of $2048 \times 8 \times 8$, while the output voxel feature ($\mathbf{RV_4}$) from Stage 4 in the point branch has a dimensional size of $256 \times 8 \times 8 \times 8$. A global average pooling operation is performed on the image features, and a new feature vector with a dimensional size of $2048 \times 1 \times 1$ is output. Simultaneously, the voxel features first undergo a dense pooling operation, which is followed by another global average pooling operation, and the output feature vector has a dimensional size of $256 \times 1 \times 1$.

The above two features are concatenated to form a 1-D vector with a size of $2304 \times 1 \times 1$, serving as an input to the classification module, which consists of a fully connected layer and a SoftMax layer. The whole process can be depicted in Fig. 8. The final outputs are probabilities that a given scene belongs to each category

$$\text{Prob} = \text{Softmax}\left(\text{FCL}\left(\text{GAP}\left(\text{DP}\left(\mathbf{RV_4}\right)\right) \oplus \text{GAP}\left(\mathbf{RI_4}\right)\right)\right) \tag{9}$$

where FCL represents the fully connected layer, GAP the global average pooling module, DP the dense pooling module, and $\oplus$ is the concatenated operation.

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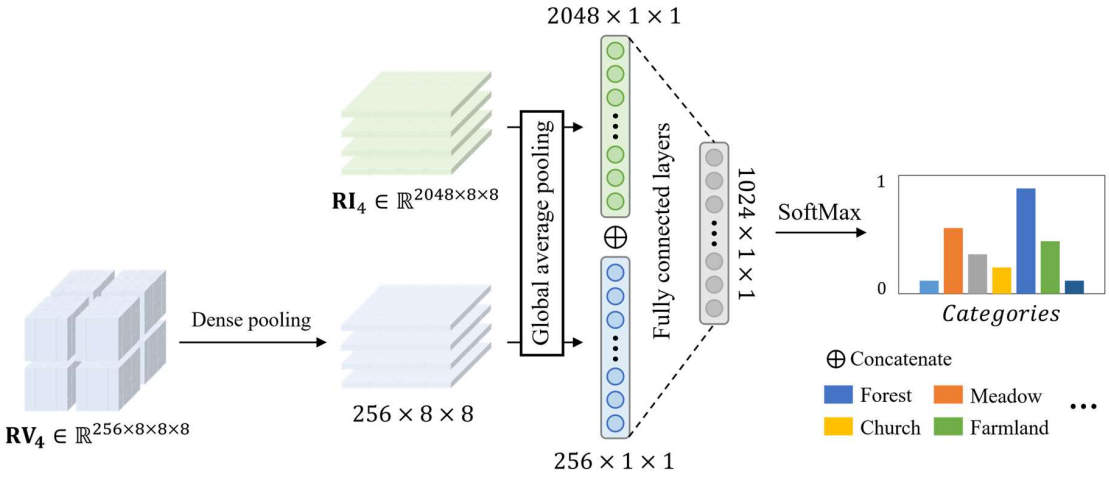


Fig. 8. Illustration of the classification module.

Optimization of the IP-SceneNet is realized by minimizing the cross-entropy loss function, expressed as follows:

$$\text{Loss} = -\sum_{i=1}^{N}\sum_{k=1}^{C} y_{i,k}\log\left(p_{i,k}\right) \tag{10}$$

where $p_{i,k}$ is the predicted probability of the ith sample belonging to the kth category, $y_{i,k}$ is the ground-truth value of the ith sample to the kth category, C is the total number of remote sensing scene categories, and N is the total number of samples.

V. Experiments and Results

In this section, we first describe the experimental settings. We then present the remote sensing scene classification results on the 2D3DScene-Netherlands and 2D3DScene-France datasets. Finally, we compare the performance of our IP-SceneNet with that of the state-of-the-art methods. The best values in Tables II and III are highlighted in bold.

A. Experimental Settings

1) Implementation Details: All experiments were executed in the Ubuntu 20.04.4 operating environment, with models being constructed using the Pytorch framework. The hardware infrastructure employed for the research consisted of one Nvidia RTX 3090 GPU in conjunction with one AMD Ryzen ThreadRipper 3990X CPU. We train the proposed IP-SceneNet from scratch in an end-to-end format. During the training phase, the Adam optimizer and cross-entropy loss function were deployed. The learning rate was stipulated at 0.0001, while the weight decay parameter was set to 0.0001, and the batch size was set to 16.

In the 2D3DScene-Netherlands and 2D3DScene-France datasets, 80% of samples were randomly selected for training and the remaining 20% for testing purposes. The number of epochs for each dataset is set to 100, and the training time for both datasets is approximately 5.0 h.

2) Evaluation Metrics: The confusion matrix serves as a well-recognized framework for classification accuracy evaluations. It is structured that the rows represent the actual categories associated with the remote sensing scenes, while the columns indicate the predicted categories. Based on the confusion matrix, two key metrics for assessing the overall performance of the proposed method were adopted: the overall accuracy (OA) and the Kappa coefficient (Kappa). Their formula can be expressed as follows:

$$\begin{cases} \text{OA} = \dfrac{\sum_{i=1}^{r} x_{ii}}{N} \\ \text{Kappa} = \dfrac{N\sum_{i=1}^{r} x_{ii} - \sum_{i=1}^{r}\left(x_{i+}\cdot x_{+i}\right)}{N^2 - \sum_{i=1}^{r}\left(x_{i+}\cdot x_{+i}\right)} \end{cases} \tag{11}$$

where x_{ii} is the value of the ith row and ith column in the confusion matrix, representing the number of correctly classified samples in the ith category; r is the number of rows or columns, which represents the total number of categories; N is the total number of samples; x_{i+} and x_{+i} the sum of elements in the ith row and the ith column, respectively.

In addition, the classification accuracy of each category is evaluated by $F1$-scores

$$F1 = \frac{2\times \text{TP}_i}{2\times \text{TP}_i + \text{FP}_i + \text{FN}_i} \tag{12}$$

where TP_i, FP_i, and FN_i are the number of true positive, false positive, and false negative samples for ith category.

B. Classification Results of Remote Sensing Scenes

Table II presents the quantitative classification results for remote sensing scenes in The Netherlands and France. Meanwhile, we compared the proposed IP-SceneNet with other methods (see Section V-C). Fig. 9 displays the confusion matrix of the classification outcomes, which provides an intuitive visualization of the model's performance across each category.

1) Results of the 2D3DScene-Netherlands Dataset: Fig. 10 presents the classification results of several typical scenes across different sample categories in the 2D3DScene-Netherlands dataset. The predicted scenes are first visually checked, and some misclassified scenes are highlighted with red boundaries in Fig. 10. The Kappa coefficient for each category, as indicated by the confusion matrix, exceeds 0.75. Furthermore, Table II shows that the OA and Kappa coefficient

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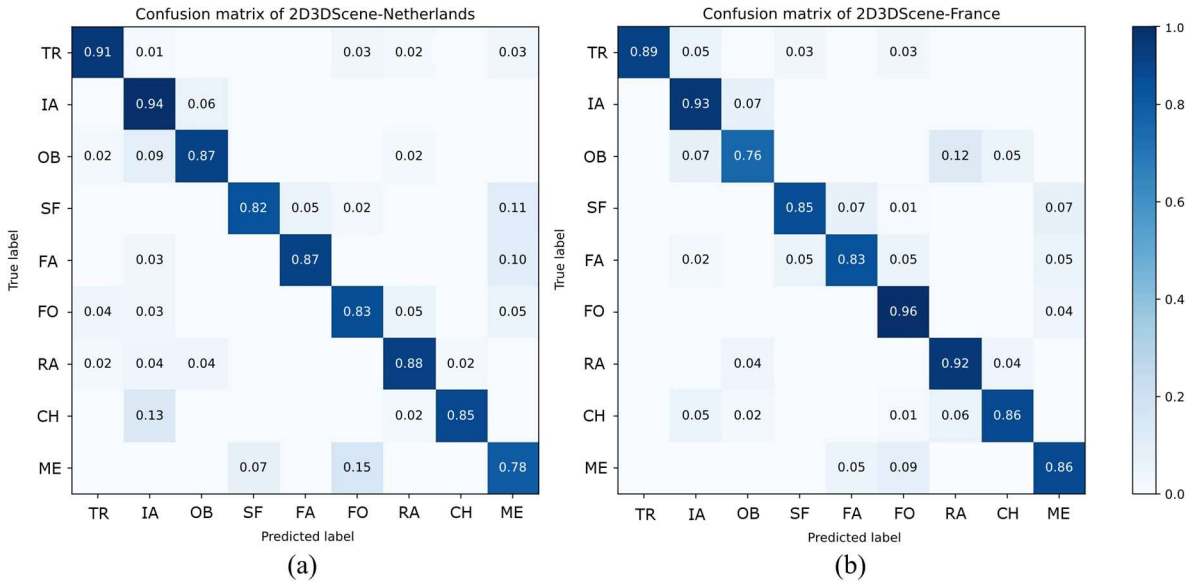


Fig. 9. Confusion matrix of the results obtained from classifying remote sensing scenes using IP-SceneNet. (a) Results obtained from classifying 2D3DScene-Netherlands. (b) Results obtained from classifying 2D3DScene-France.

TABLE II
QUANTITATIVE CLASSIFICATION RESULTS OF 2D3DSCENE-NETHERLANDS AND 2D3DSCENE-FRANCE DATASETS

Datasets	F1 (%)									OA (%)	Kappa
	TR	IA	OB	SF	FA	FO	RA	CH	ME		
2D3DScene-Netherlands	**91.46**	82.82	88.32	86.77	90.63	81.77	88.44	90.91	75.36	86.11	0.8438
2D3DScene-France	**94.18**	87.74	80.42	88.08	85.13	89.30	87.62	88.21	85.15	87.33	0.8575

for IP-SceneNet are 86.11% and 84.38%, respectively, further demonstrating the effectiveness of the proposed method.

In terms of individual category, the best performance was obtained in the category of TR, with an $F1$-score of 91.46%. The CH category had the second-highest $F1$-score of 90.91%. The lowest $F1$-score was 75.36%, which was achieved in the category of ME. This is because there are many pastures in The Netherlands, which are agricultural land and similar to MEs, leading to easy misclassification. The third to the second last $F1$-scores were 81.77% and 82.82%, which were associated with categories of FO and IA. These results demonstrated that our proposed IP-SceneNet can successfully classify most of the categories with high accuracy.

According to the confusion matrix in Fig. 9(a), misclassifications mainly exist among IA, OB, and RA, as well as among ME, FA, and FO. Fig. 11 lists several typical misclassification cases in the 2D3DScene-Netherlands dataset. Row 1 exemplifies several remote sensing scenes belonging to OB that are incorrectly classified as IA. This confusion mainly stems from their similar appearances and geometrical structures of large footprints and low height. Furthermore, some sports fields were noticed to be erroneously identified as FA, FO, or ME, as shown in row 2 of Fig. 11. This misclassification can be attributed to that some sports fields are comprised of green landscapes without clear boundaries. In row 3 of Fig. 11, misclassifications happened among FA, FO, and ME. Such confusion is mainly caused by their analogous spectral characteristics as they all belong to vegetation. In addition, the prevalence of pastoral landscapes in The Netherlands exacerbates the challenges of distinguishing these categories. In row 4 of Fig. 11, several CHs are misclassified as RAs or IAs. By carefully checking these CHs, we find that they have conventional architectural styles, such as flat roofs, which lack distinctive religious characteristics. As far as we know, some CHs even directly rent abandoned factory buildings or empty residential houses as their gathering places, which can also lead to confusion.

2) Results of the 2D3DScene-France Dataset: Fig. 12 illustrates the classification results of typical scenes in the 2D3DScene-France dataset. Similarly, the misclassified scenes are highlighted with red boundaries. The quantitative classification performance of this dataset is also detailed in Table II. The OA and Kappa coefficient of the 2D3DScene-France dataset reach 87.33% and 85.75%, which are both slightly higher than those of the 2D3DScene-Netherlands dataset.

In terms of individual categories, the TR category exhibits the highest $F1$-score of 94.18%. The lowest $F1$-score was achieved by the OB (80.42%). FA and ME have the third to the second last $F1$-scores, which were 85.15% and 85.13%. The above results again validated that IP-SceneNet performed well on the 2D3DScene-France dataset and obtained satisfactory accuracies on most of the categories.

The confusion matrix of the 2D3DScene-France dataset was depicted in Fig. 9(b), showing that the main misclassification

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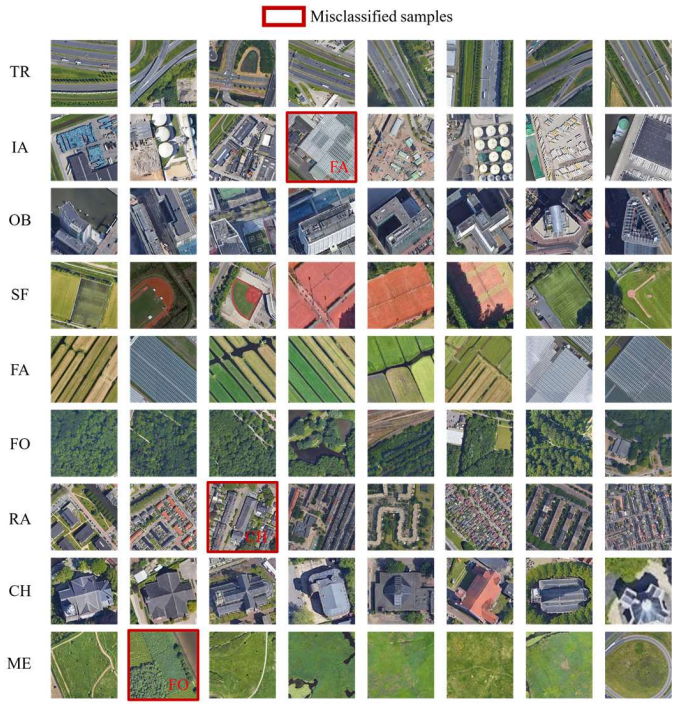


Fig. 10. Classification results of the 2D3DScene-Netherlands dataset by category using IP-SceneNet. Labels in the first column are the predicted ones, and samples enclosed with red boundaries are misclassified, with the actual label annotated on the image.

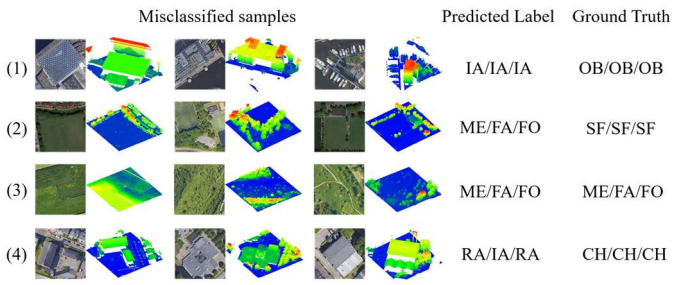


Fig. 11. Examples of misclassified samples in the classification results on the 2D3DScene-Netherlands dataset.

happened among ME, FA, IA, and RA. Furthermore, a selection of representative misclassified samples is showcased in Fig. 13. In row 1 of Fig. 13, some scenes belonging to IA were incorrectly classified as the TR category. This is due to the predominance of cement flooring in both IAs and TR fields. In row 2 of Fig. 13, OBs were erroneously categorized as IA or RA. In some French cities, like Lyon, Marseille, and Montpellier, OBs are scarce, and the existing OBs lack unique structures, which may explain the misclassification among OBs, factories, and residential buildings. Moreover, it is worth mentioning that the classification performance of the FA category ($F1$: 85.13%) in the 2D3DScene-France dataset falls short compared to that of the 2D3DScene-Netherlands dataset ($F1$: 90.63%). Such a disparity can be attributed to that the HSR images in the 2D3DScene-France dataset are dimmer and cloudier (row 3 of Fig. 13) than those in the 2D3DScene-France dataset, making it more difficult to distinguish boundaries between farm fields. However, the clarity of the boundaries is a pivotal characteristic for distinguishing

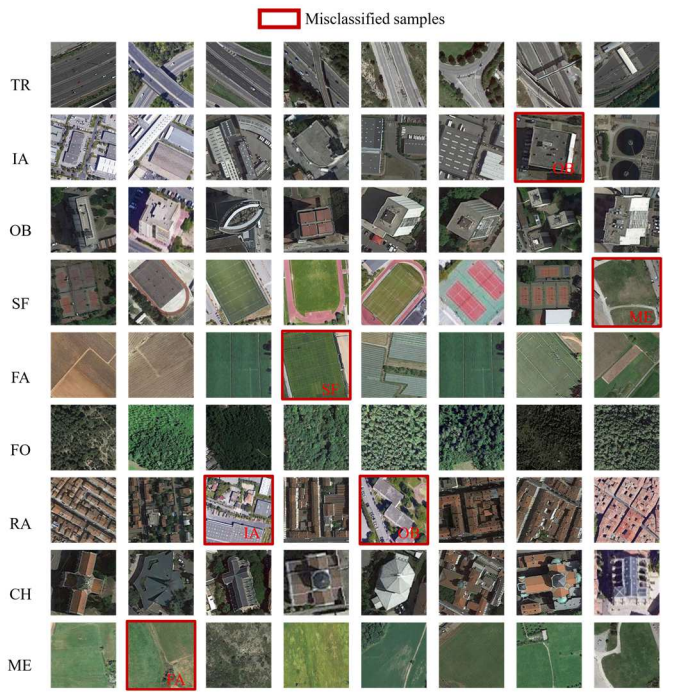


Fig. 12. Classification results of the 2D3DScene-France dataset by category using IP-SceneNet. Labels in the first column are the predicted ones, and samples enclosed with red boundaries are misclassified, with the actual label annotated on the image.

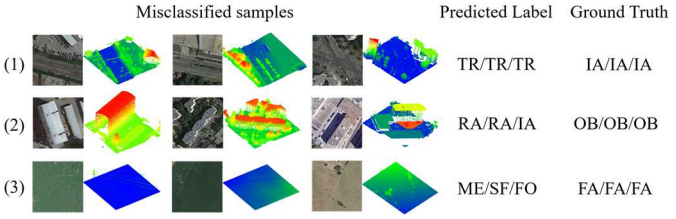


Fig. 13. Examples of misclassified samples in the classification results on the 2D3DScene-France dataset.

FA from other vegetation categories, including ME and FO, adding additional challenges for scene classification.

C. Comparisons With Other Existing Methods

In order to evaluate the effectiveness of the proposed IP-SceneNet, we selected several existing methods and categorized them into three groups for comparative experiments: pure 3-D networks, pure 2-D networks, and 2-D–3-D multimodal networks. We prioritized selecting methods that: 1) are well-recognized in the literature; 2) provide open-source implementations for fair comparison; and 3) support multimodal inputs. Specifically, for the pure 3-D networks, three methods were selected: PointNet [61], PointNet++ [73], and RandLA-Net [74]. Given the large scale of point cloud data in remote sensing scenes, the farthest point sampling (FPS) method was applied to downsample the point clouds of all samples to 30 000 points prior to feature extraction. Subsequently, the random sampling was employed during the downsampling stage of the feature extraction process. The pure 2-D networks include GCSANet [75], ConvNeXt [76], and ACNet [77]. These methods represent

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TABLE III
QUANTITATIVE RESULTS OF COMPARISON EXPERIMENTS IN 2D3DSCENE-NETHERLANDS

Type	Methods	F1 (%)									OA (%)	Kappa
		TR	IA	OB	SF	FA	FO	RA	CH	ME		
2D	GCSANet	87.32	81.22	68.53	82.35	83.63	75.23	87.55	69.57	70.13	82.36	0.7990
	ConvNeXt	85.54	80.19	73.85	80.26	82.34	76.63	77.83	72.37	73.25	81.73	0.7886
	ACNet	87.63	75.29	76.42	88.65	84.67	78.52	78.92	75.67	73.78	82.59	0.8049
3D	PointNet	65.35	73.33	75.32	42.92	45.69	71.13	74.85	72.56	53.65	61.73	0.5729
	PointNet++	78.62	83.50	82.83	68.91	58.70	73.49	76.26	71.44	53.40	75.43	0.7212
	RandLA-Net	85.35	83.29	86.06	72.45	68.93	72.73	82.74	78.58	58.96	80.51	0.7852
2D-3D	PIF-Net	73.29	**85.33**	88.23	79.35	78.54	75.63	88.96	**91.03**	74.46	83.88	0.8207
	CMGFNet	88.76	84.39	**90.51**	82.16	85.23	80.54	88.01	90.11	69.19	84.56	0.8219
	RF	85.18	69.78	84.62	80.57	86.93	68.85	84.88	81.33	66.62	76.89	0.7354
	KNN	83.83	74.65	84.00	78.97	85.29	72.37	83.41	83.43	64.96	75.50	0.7263
	LDA	82.78	70.16	80.32	76.55	83.21	69.59	73.32	80.87	62.54	73.68	0.7095
	IDF-CNN	88.20	81.78	84.34	**87.36**	85.11	78.30	87.72	81.09	73.15	81.52	0.7913
	IP-SceneNet	**91.46**	82.82	88.32	86.77	**90.63**	**81.77**	**88.44**	90.91	**75.36**	**86.11**	**0.8438**

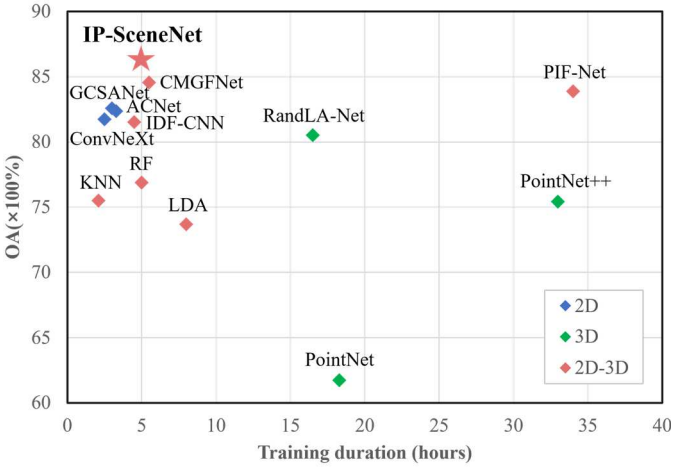


Fig. 14. Comparison of training accuracy and training duration of different classification methods. Blue, green, and red represent 2-D, 3-D, and 2-D–3-D multimodal networks, respectively.

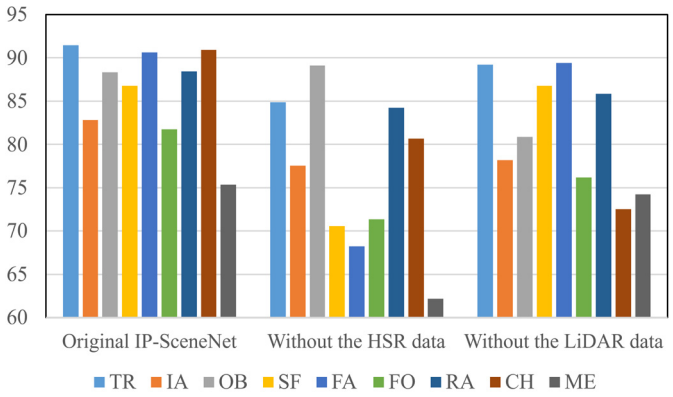


Fig. 15. $F1$-scores of each category in the three experiments.

state-of-the-art approaches for classifying remote sensing scenes based solely on remote sensing images. The 2-D–3-D multimodal network group encompasses six methods: RF, KNN, and LDA [31], IDF-CNN [32], PIF-Net [67], and CMGFNet [58]. RF, KNN, and LDA were utilized by Sanlang et al. [31] to extract spectral and texture features from HSR images and 3-D geometric features (e.g., height, sky view factor, DSM, and flatness) from LiDAR point clouds. These extracted 2-D and 3-D features were then input into RF, KNN, and LDA classifiers, respectively. Both IDF-CNN and CMGFNet convert point clouds into DSM images and subsequently fuse them with HSR images. PIF-Net directly extracts point cloud features and performs hierarchical fusion with image features. To ensure the fairness of the comparative experiments, the selected approaches were systematically applied to the 2D3DScene-Netherlands datasets. Their results were meticulously analyzed and quantitatively compared, as delineated in Table III.

According to Table III, our proposed IP-SceneNet achieved the highest OA and Kappa coefficient in the 2D3DScene-Netherlands, which was 1.55% and 2.19% higher than the second highest achieved by CMGFNet. In terms of individual category, IP-SceneNet obtained the highest $F1$-score in most of the categories, i.e., TR, FA, FO, RA, and ME. The above results fully validated the effectiveness and superiority of our proposed IP-SceneNet. Furthermore, among different methodological categories, approaches that integrate both 2-D and 3-D information consistently demonstrate superior classification performance. This observation indicates that the integration of complementary 2-D and 3-D features effectively enhances remote sensing scene classification to a significant degree. In addition, purely 2-D-based methods generally outperform their purely 3-D-based counterparts. This disparity primarily stems from the tendency of pure 3-D methods to employ excessive downsampling, which induces substantial information loss and consequently detrimentally impacts the classification accuracy.

Furthermore, we compared the proposed IP-SceneNet with existing 2-D, 3-D, and 2-D–3-D multimodal networks in terms of both training duration and OA. The corresponding results are presented in Fig. 14. Notably, the time of FPS downsampling was not counted in pure 3-D approaches, while the time of feature extraction was not counted for RF, LDA, and KNN. As illustrated in Fig. 14, it took about 5.0 h to train the IP-SceneNet, and an OA of 86.11% was achieved. Compared with other models, a notable balance between classification accuracy and training efficiency for remote sensing scene classification was achieved by the IP-SceneNet. Among all these

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TABLE IV
QUANTITATIVE RESULTS OF REMOTE SENSING SCENE CLASSIFICATION AFTER DISCARDING DIFFERENT TYPES OF DATA

Datasets	F1 (%)									OA (%)	Kappa
	TR	IA	OB	SF	FA	FO	RA	CH	ME		
Original IP-SceneNet	**91.46**	**82.82**	88.32	86.77	**90.63**	**81.77**	**88.44**	**90.91**	**75.36**	**86.11**	**0.8438**
Without the HSR data	84.85	77.54	**89.12**	70.57	68.24	71.35	84.22	80.69	62.19	75.00	0.7213
Without the LiDAR data	89.22	78.18	80.85	**86.79**	89.43	75.16	85.83	70.54	72.25	81.73	0.7890

TABLE V
RESULTS OF PROCESSING THE DENSE POOLING MODULE UNDER DIFFERENT CONDITIONS

Conditions	OA (%)	Kappa
Original IP-SceneNet	**86.11**	**0.8438**
Without the dense pooling module	83.07	0.8135
X-axis max pooling	84.21	0.8249
Y-axis max pooling	85.29	0.8346
Z-axis average pooling	85.42	0.8365

TABLE VI
COMPARATIVE RESULTS WITH/WITHOUT THE IVA MODULE

Conditions	OA (%)	Kappa
Original IP-SceneNet	**86.11**	**0.8438**
Without the IVA module	85.03	0.8312

TABLE VII
QUANTITATIVE CLASSIFICATION RESULTS CROSS-VALIDATED ON DATASETS 2D3DSCENE-NETHERLANDS AND 2D3DSCENE-FRANCE

Conditions	OA (%)	Kappa
NE on NE	86.11	0.8438
FE on FE	87.33	0.8575
NE on FE	84.63	0.8222
FE on NE	83.23	0.8159

NE represents the 2D3DSCENE-NETHERLANDS dataset, FE represents the 2D3DSCENE-FRANCE dataset.

networks, the IP-SceneNet attained the highest classification accuracy while maintaining a moderate training duration.

VI. DISCUSSION

In this section, we first discuss the respective contributions of HSR and LiDAR data to remote sensing scene classification. We then conduct ablation studies to validate the necessity of each component within the proposed IP-SceneNet model.

A. Contributions of HSR Images and LiDAR Data

This article utilizes HSR images in conjunction with LiDAR data to extract both 2-D spectral features and 3-D structural characteristics. To investigate their distinct contributions to scene classification, we conducted two ablation experiments on the 2D3D-Netherlands dataset. Specifically, the experiments involved selectively removing one type of data while retaining the other for analysis. The outcomes of each ablation experiment and the original study were presented in Table IV for comparative evaluations. In addition, the $F1$-scores of each category in three experiments were plotted in Fig. 15.

As depicted in Table IV, the removal of either the HSR images or the LiDAR data results in a significant decrease in overall classification accuracy. After removing HSR images, OA and Kappa coefficient are decreased by 9.11% and 10.25%, respectively. Meanwhile, a decrease of 5.33% and 6.00% in OA and Kappa coefficient is observed after removing the LiDAR data. These results indicate that both HSR images and LiDAR data play an important role in providing distinctive clues and improving the accuracy of scene classification. The decrease in accuracy after removing HSR images is larger than that of removing the LiDAR data, demonstrating that 2-D spectral features are the key factors in distinguishing different scenes, while 3-D structural characteristics embedded in the LiDAR data are powerful supplements.

Fig. 15 illustrates the impact of excluding either of the two data types on the classification accuracies of different categories. According to Fig. 15 and Table IV, a significant decrease in $F1$ was observed after removing either the HSR images or the LiDAR data for most categories. However, the exclusive use of LiDAR data achieves nearly the same results

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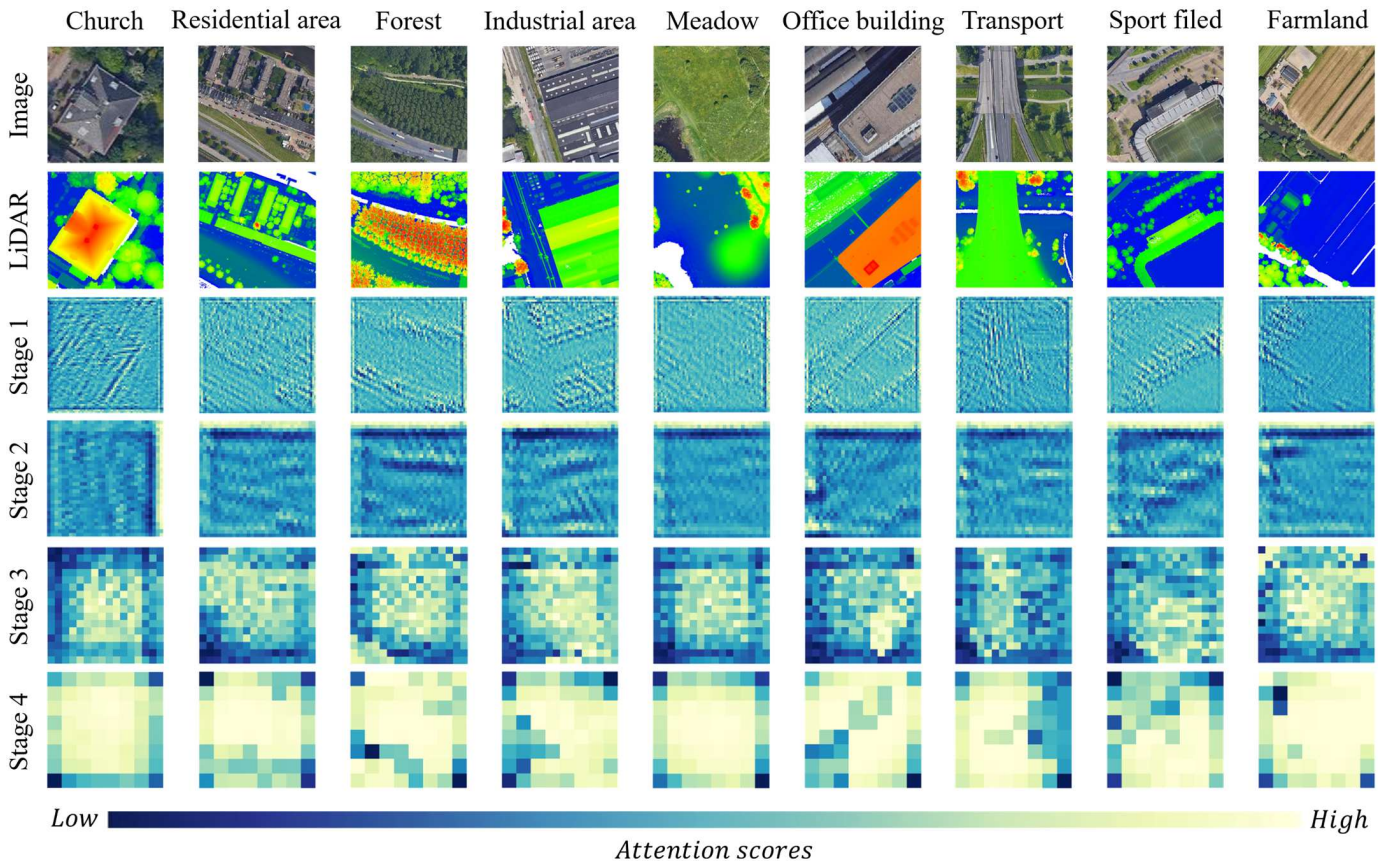


Fig. 16. Visualizations of attention weight distribution map in Section IV-D. First row: HSR images of different scene categories. Second row: LiDAR point clouds (top views) of different scene categories. Third to sixth rows: Attention maps in stage 1–stage 4.

as fusion for SF and FA categories, whereas the exclusive use of HSR images yields comparable performance for the OB category. These results suggest that in some categories, single-modal approaches already achieve the strong performance. However, in some challenging cases where structural and spectral cues complement each other, the fusion of HSR images and LiDAR data yields notable improvements.

B. Necessities of the Components in IP-SceneNet

IP-SceneNet comprises two fundamental components: dense pooling module and IVA module. To ascertain the efficacy of these modules, additional experiments were carried out on the 2D3D-Netherlands dataset by systematically removing or modifying each of the two modules.

1) Efficacy of the Dense Pooling Module: This module mitigates the sparsity of voxel features, enabling more balanced and informative fusion with dense image features. To evaluate the effectiveness of this module, four supplementary experiments were executed. The first experiment directly removed the dense pooling module, while the second and third experiments modified the pooling direction from the top-down view (Z-axis) to side-view perspectives, i.e., the X- and Y-axis directions. The fourth experiment modified the pooling type from maximum pooling to average pooling. The results of these experiments were evaluated and compared against the original model, as detailed in Table V.

According to Table V, it was noted that a decrease of 3.04% and 3.03% in the Kappa coefficient was observed when the dense pooling module was eliminated. Such a decrease underscores the pivotal role of the dense pooling module in fusing multimodal features and improving the performance of IP-SceneNet. Notably, altering the pooling direction to side-view orientations resulted in diminished accuracies compared with the top-down perspective. Such a diminishment can be attributed to the less adeptness of side-view perspectives in averting occlusions, thereby losing original information embedded in the LiDAR data. However, the OAs and Kappa coefficients of both side-view orientations, i.e., the X- and Y-axis directions, are still higher than those without the dense pooling module, further validating the effectiveness of the pooling operation.

By replacing the maximum pooling in the dense pooling module with an average pooling operation, it was found that the OA decreased by 0.69% in OA, while the Kappa coefficient decreased by 0.73%. This decline may be explained by that maximum pooling accentuates the most salient features, while such a highlighting mechanism is inadequate in the average pooling, resulting in a slight drop in accuracy metrics.

The Z-axis max-pooling strategy effectively collapses the vertical features into a compressed form. This not only suppresses less informative or irrelevant features but also significantly reduces memory usage and computational cost, especially important given that a large proportion of voxels

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are empty. However, this strategy also introduces a tradeoff. It discards fine-grained vertical details, potentially leading to a loss of information and compromising the model's ability to capture certain vertical nuances.

2) Efficacy of the IVA Module: The IVA module allows each modality to dynamically attend to the most relevant features from the other modality. This enhances cross-modal contextual understanding and semantic alignment, which is critical for complex scene classification. To assess the efficacy of the IVA module, an additional experiment was conducted by dismantling this module. The results of the experiment were evaluated and compared with the original model, which is presented in Table VI.

According to Table VI, after removing the IVA module, the OA and Kappa coefficient were decreased to 85.03% and 83.12%, which were 1.08% and 1.26% lower than those of the original IP-SceneNet model. Such a decrease underscores the effectiveness and contributions of the IVA module to multimodal scene classifications.

Moreover, after the multimodal features pass the IVA module, an attention weight distribution map is produced. To further prove the effectiveness of the IVA module, samples are chosen from each category, and their attention weight distribution maps in different stages are visualized in Fig. 16.

The 3rd–6th rows of Fig. 16 show the attention weight distribution maps at different stages. As the stage increases, scores where main targets (e.g., trees, roads, and factories) are located grow higher and higher, and these targets in different scenes become clearer. The IVA module gradually identifies foregrounds within each sample by allocating higher attention weight scores, thereby enhancing the model's classification capabilities.

Together, the dense pooling and the IVA modules provide both practical utility by improving the classification accuracy and algorithmic novelty through a tailored mechanism for fusing highly heterogeneous data sources.

C. Generalization Capability of IP-SceneNet

To assess the generalization performance of the proposed model, particularly its ability to classify unseen geographical areas, this study conducted tests on the two datasets. The model trained on the 2D3DScene-Netherlands dataset's training set was tested on the 2D3DScene-France dataset's testing set, and vice versa. The results were compared with those presented in Sections V-B1 and V-B2. These findings are summarized in Table VII. As shown in the third and fourth rows of Table VII, IP SceneNet achieved high classification accuracy on unfamiliar geographical scenes, indicating that the proposed model exhibits strong generalization capabilities.

In addition, there is a great potential for IP-SceneNet to be transferred to other study areas and even other LiDAR sensors, including LiDAR spaceborne sensors. As spaceborne LiDAR data are more accessible than airborne LiDAR data and cover larger areas than airborne LiDAR data, they will bring more real-world applications. However, due to the limitation of space, such explorations can only be conducted in future studies.

VII. Conclusion

This study presents a multimodal framework for remote sensing scene classification using HSR images and LiDAR point clouds. Our primary contributions are twofold: 1) establishment of two benchmark datasets, i.e., 2D3DScene-Netherlands and 2D3DScene-France, that enable standardized evaluation of multimodal scene classification methods and 2) development of IP-SceneNet, an end-to-end network that effectively addresses three critical challenges in cross-modal fusion, i.e., information scale imbalance through voxelization and sparse convolution, feature density disparity via dense pooling module, and intermodality redundancy with the IVA attention mechanism. Comprehensive empirical validations have shown that our framework outperforms both traditional methods and existing deep learning approaches on both datasets.

Experimental evaluations on the 2D3DScene-Netherlands and 2D3DScene-France datasets demonstrate that IP-SceneNet ranks first among the purely 3-D networks, purely 2-D networks, and 2-D–3-D multimodal networks. Ablation studies have been conducted, and two conclusions can be drawn: 1) both HSR imagery and LiDAR data are crucial for enhancing scene classification accuracy, with HSR imagery contributing more significantly to the results and 2) the dense pooling module and the IVA module are indispensable to the superior performance of the IP-SceneNet.

The current limitations mainly stem from a reliance on high-quality labeled samples. Future work will focus on: 1) incorporating domain knowledge to minimize reliance on labeled data and 2) broadening the dataset in terms of scale and geographic coverage.

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